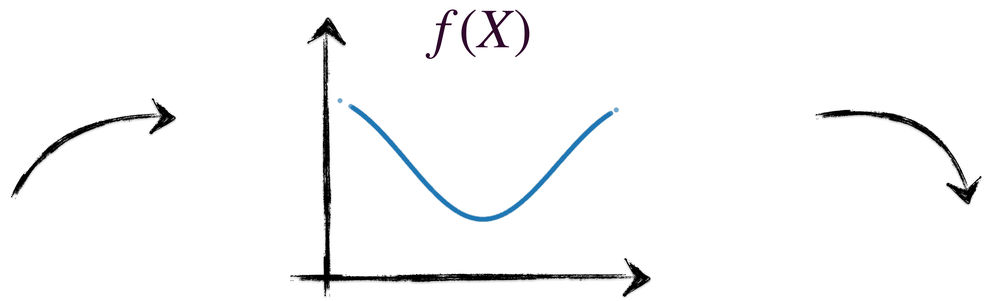
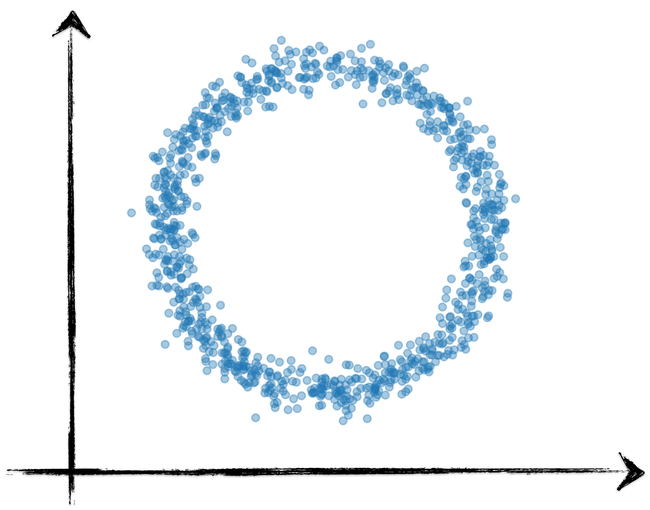
How Do We Capture Nonlinear Statistical Dependency ?



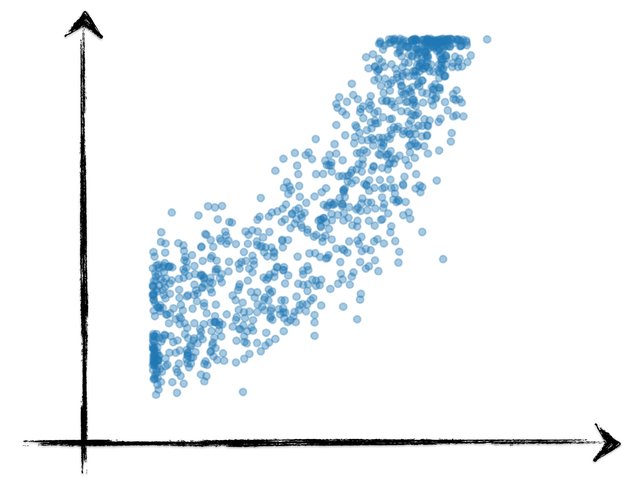
Correlation = 0.03

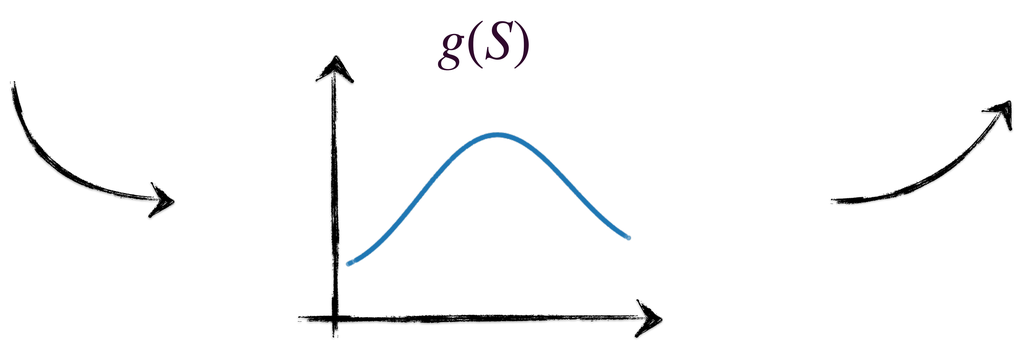


Transform Nonlinear Dependency between X and S

Linear Dependency between $f(X)$ and $g(S)$

Correlation = 0.88





How Do We Formalize a Nonlinear Statistical Dependency Test ?

Let $\mathscr{F}$ and $\mathscr{G}$ be a characteristic RKHS